

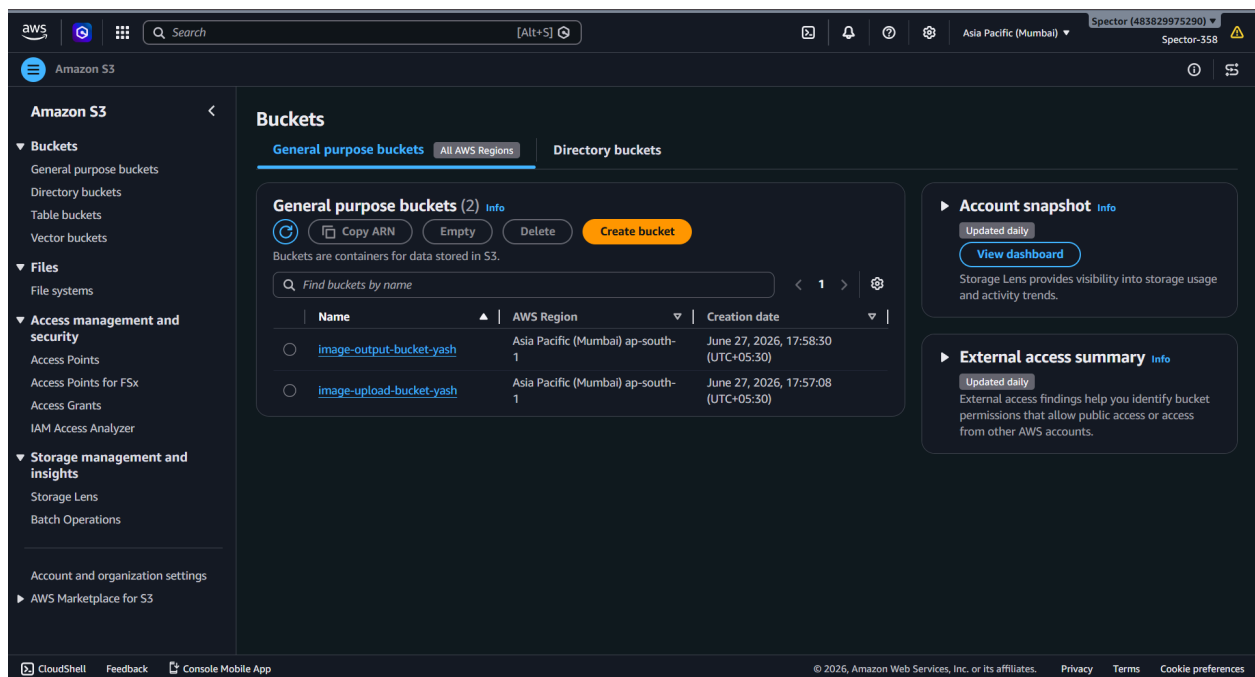
# Serverless Image Processing Platform

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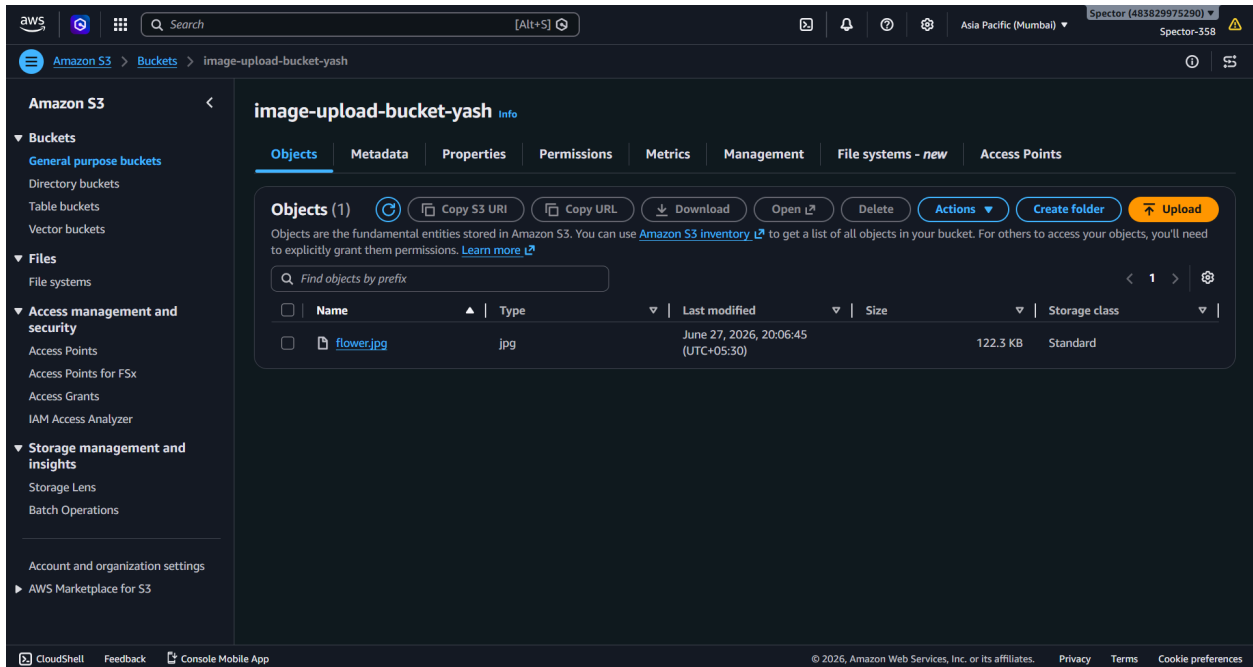
**Problem Statement:** A company needs an image processing platform that can scale dynamically based on demand. The existing on-premises solution struggles with scalability during peak times.

**This document contains all the project screenshots.**



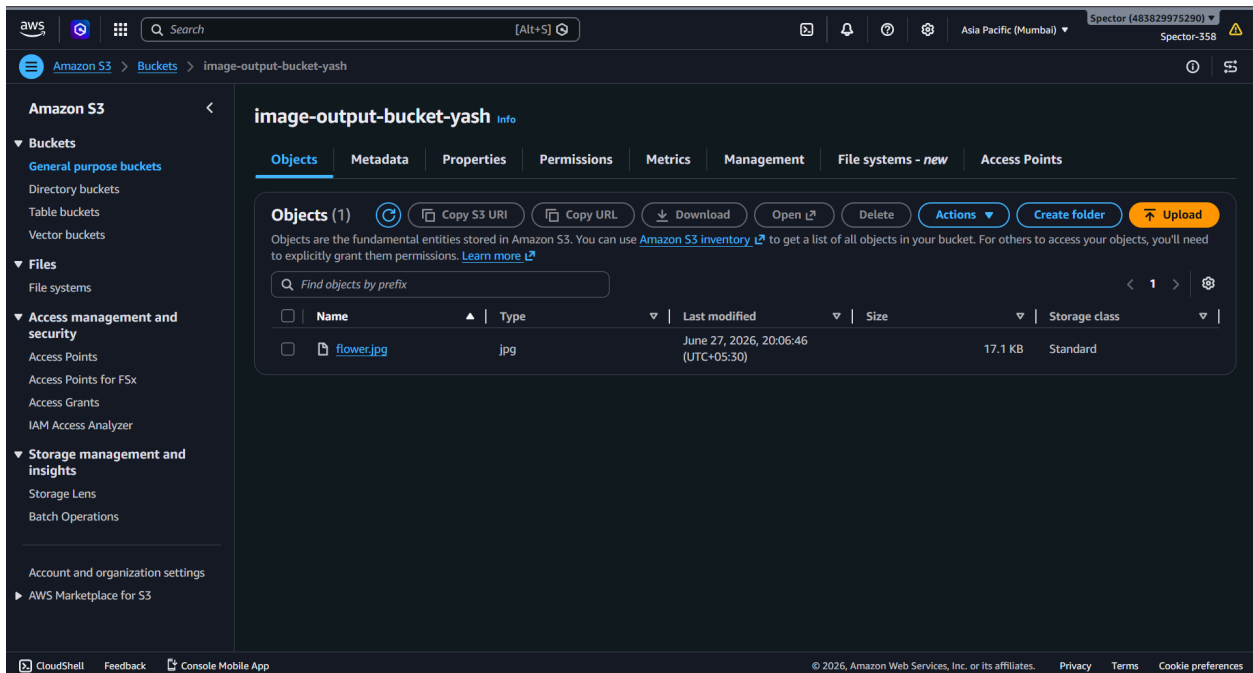
*Fig. 1 : Amazon S3 Buckets*

Shows the two S3 buckets used in the project: one for uploading original images and another for storing processed images.



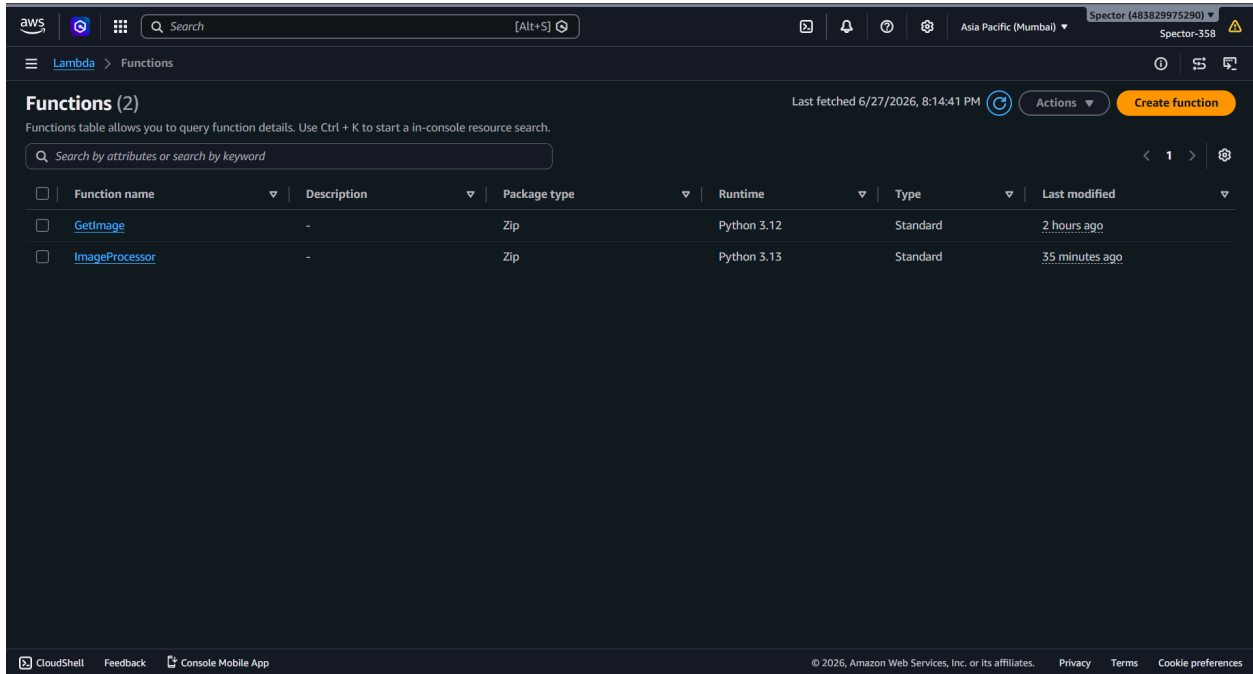
*Fig. 2 : Image Upload Bucket*

Demonstrates that the original image (flower.jpg) is successfully uploaded to the input S3 bucket.



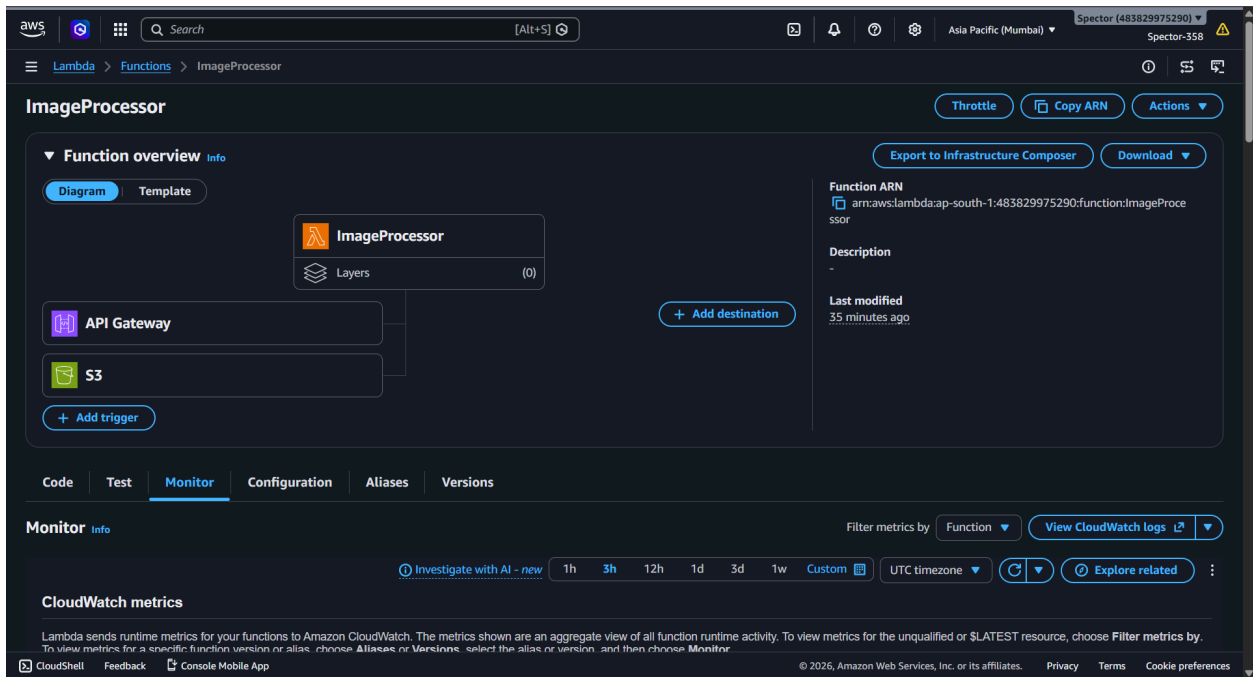
*Fig. 3 : Processed Image Bucket*

Shows that the resized image is automatically stored in the output bucket after Lambda processing.



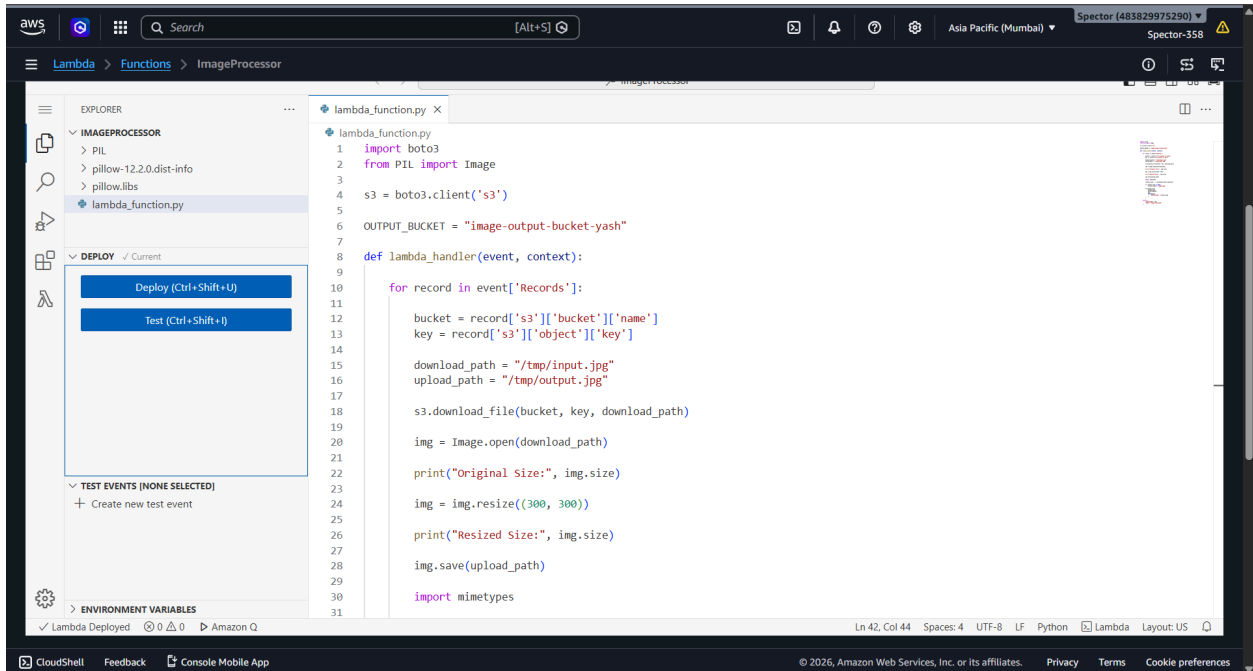
*Fig. 4 : AWS Lambda Functions*

Displays the two Lambda functions (ImageProcessor and GetImage) used for image processing and image retrieval.



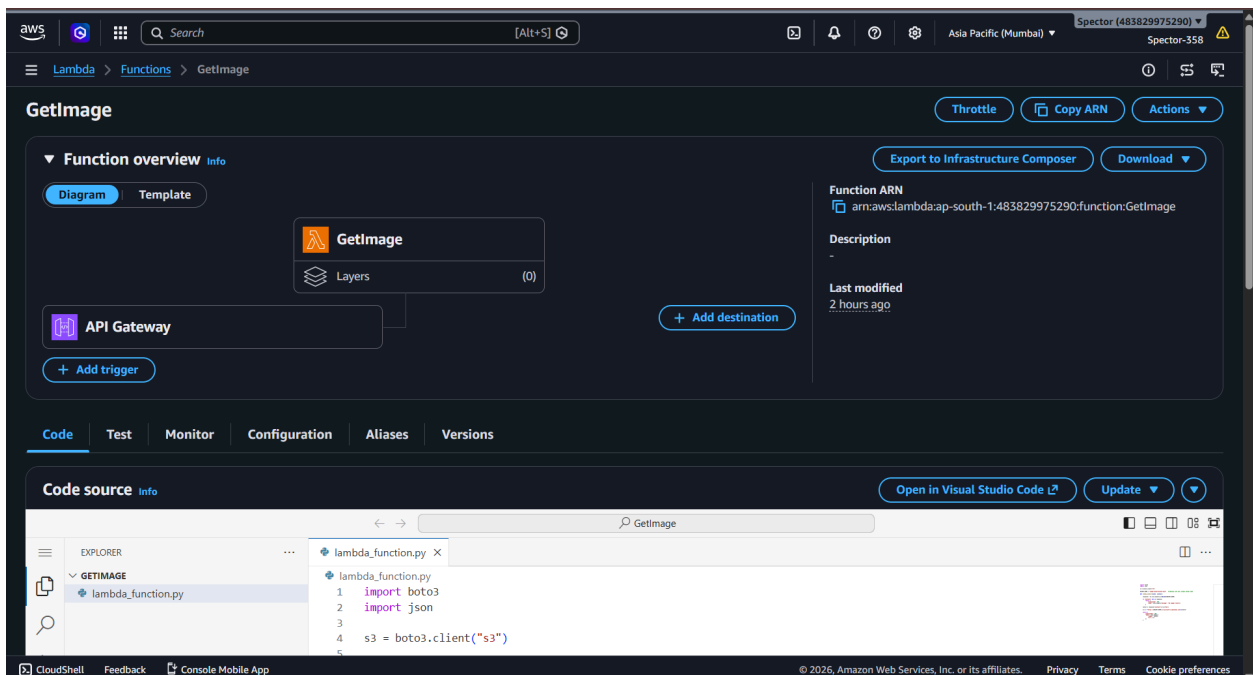
*Fig. 5 : ImageProcessor Lambda Overview*

Shows the architecture of the ImageProcessor Lambda function with S3 and API Gateway integrations.



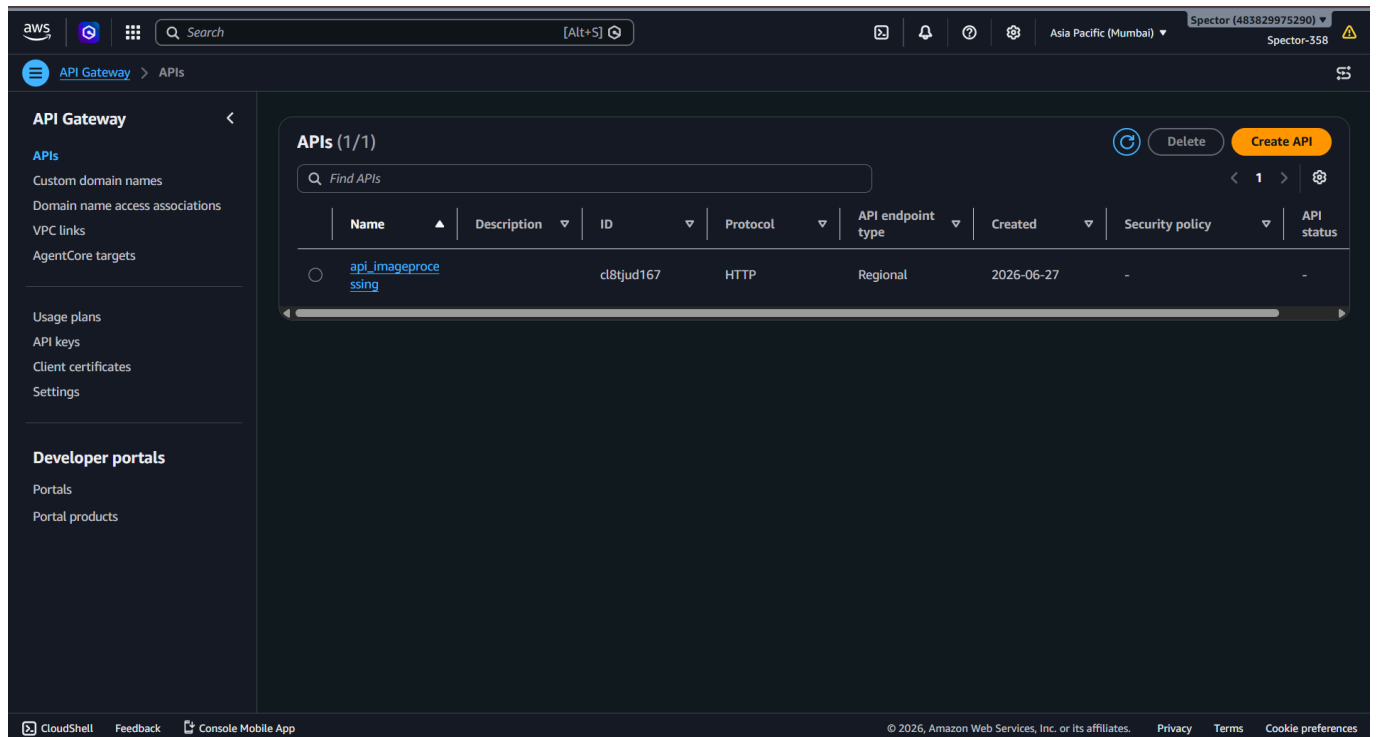
*Fig. 6 : ImageProcessor Lambda Source Code*

Displays the Python code responsible for downloading, resizing, and uploading the processed image.



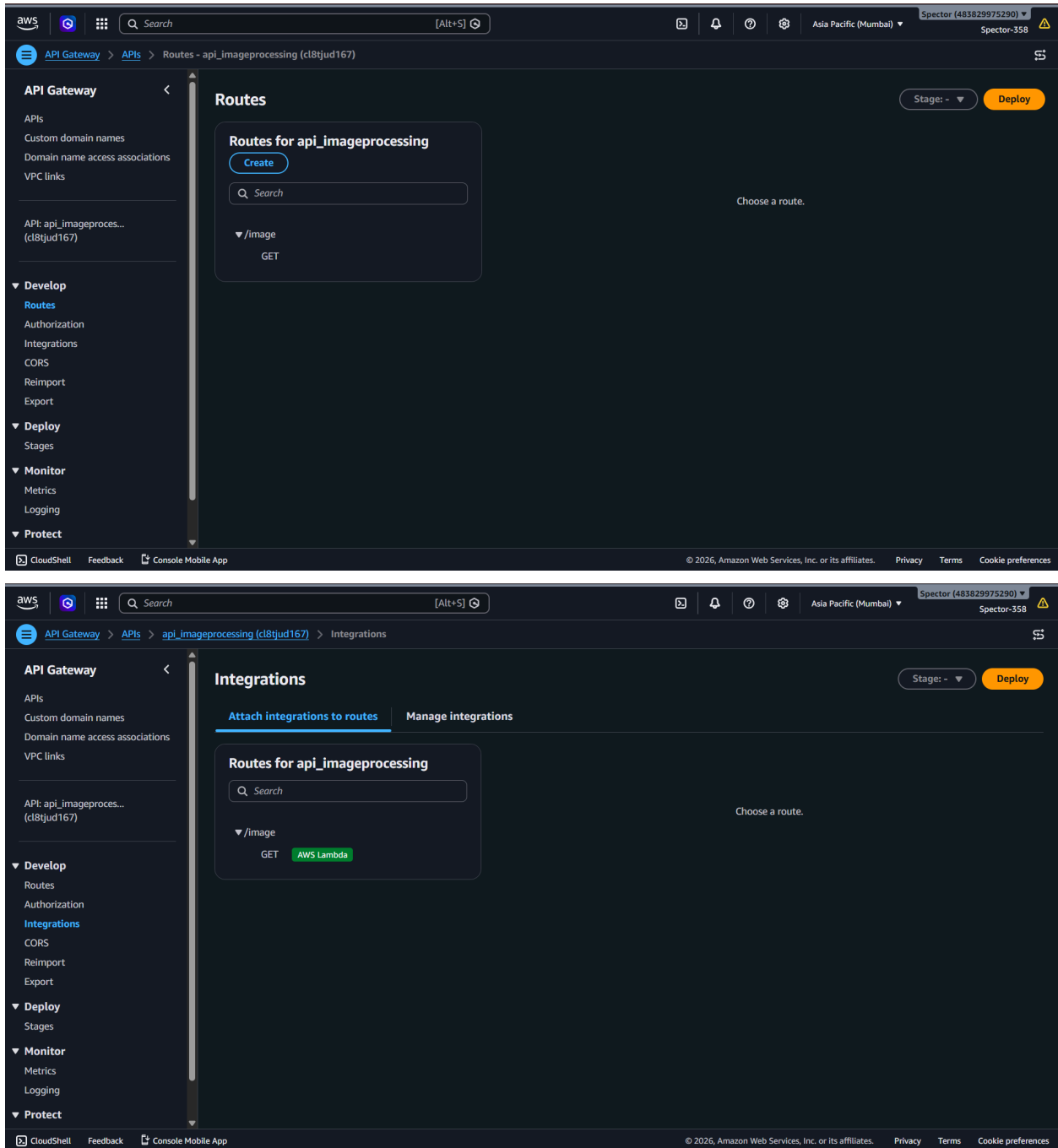
*Fig. 7 : GetImage Lambda Function*

Shows the Lambda function that retrieves the processed image URL from the output bucket.



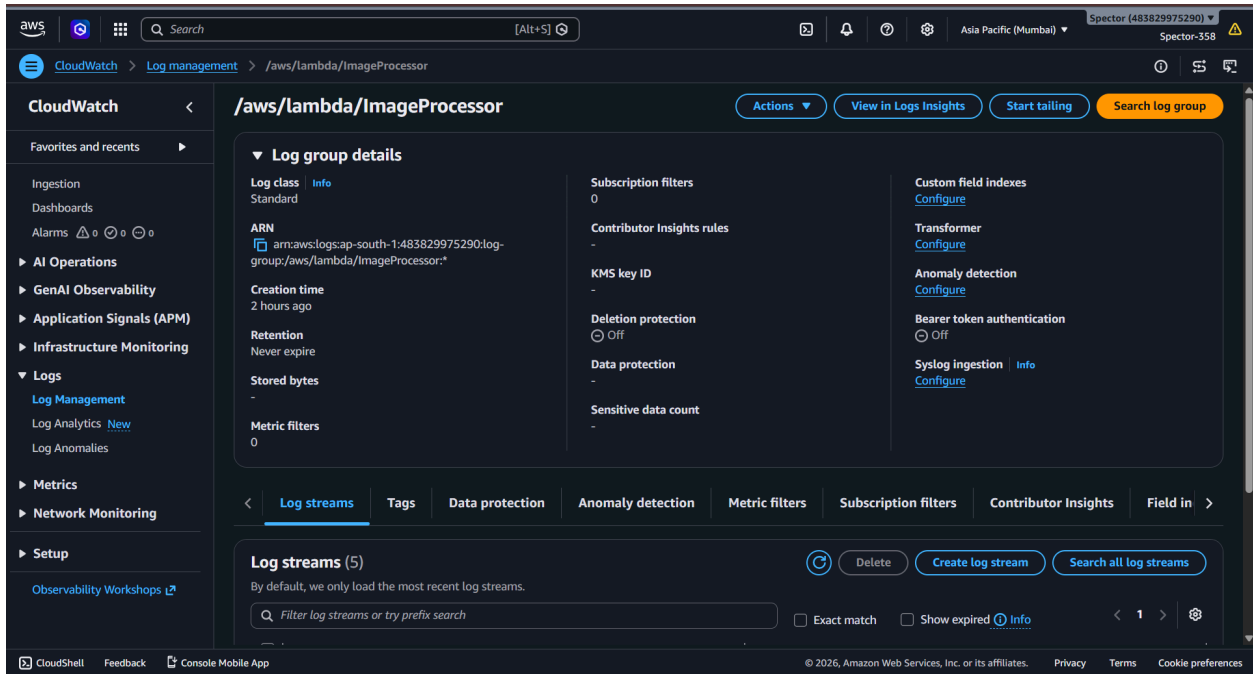
*Fig. 8 : API Gateway Configuration*

Displays the HTTP API created to allow users to access the processed image through an endpoint.

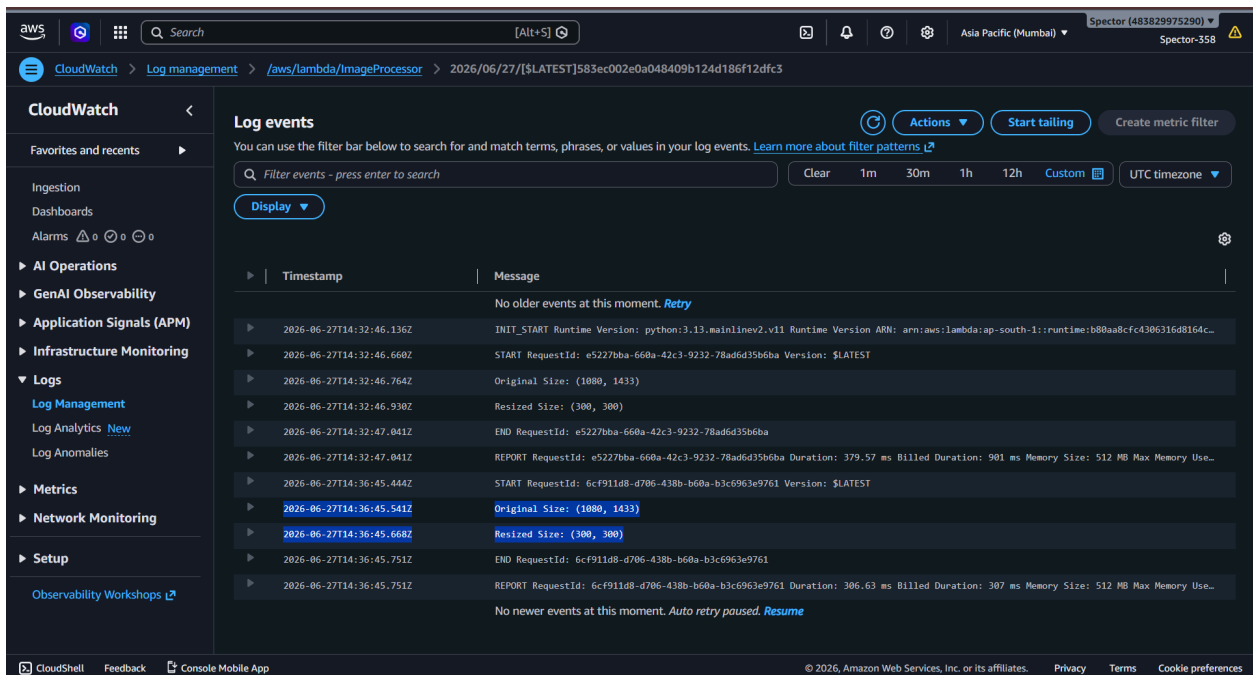


*Fig. 9 : API Gateway Route and Integration*

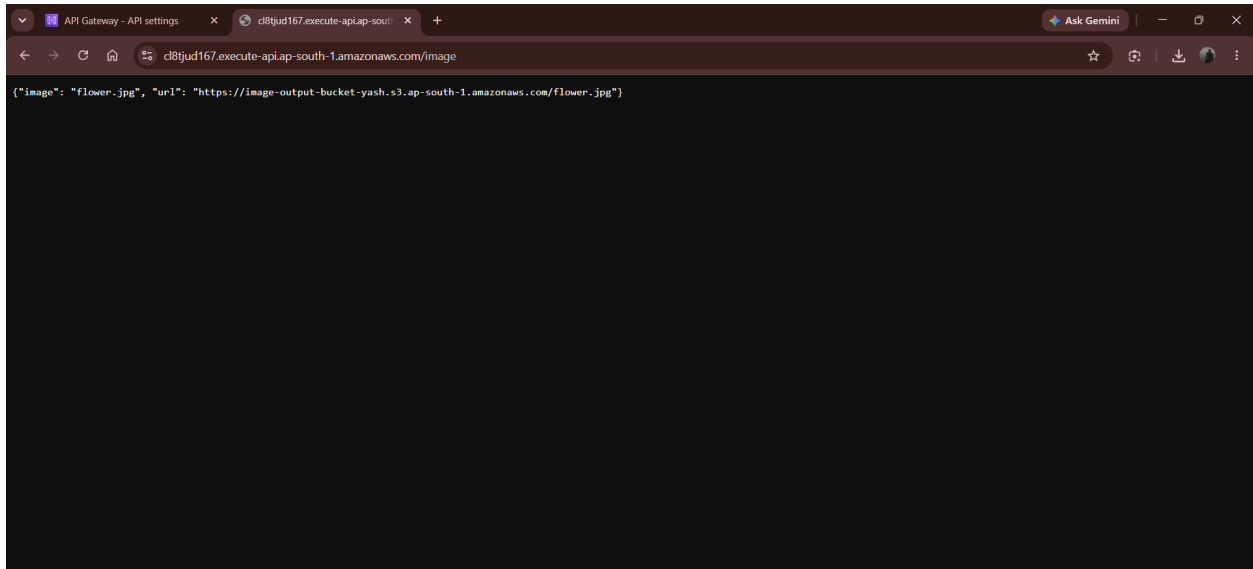
Shows the GET `/image` route and its integration with the `GetImage` Lambda function.



*Fig. 10 : CloudWatch Log Group*  
Demonstrates CloudWatch log monitoring for Lambda execution.

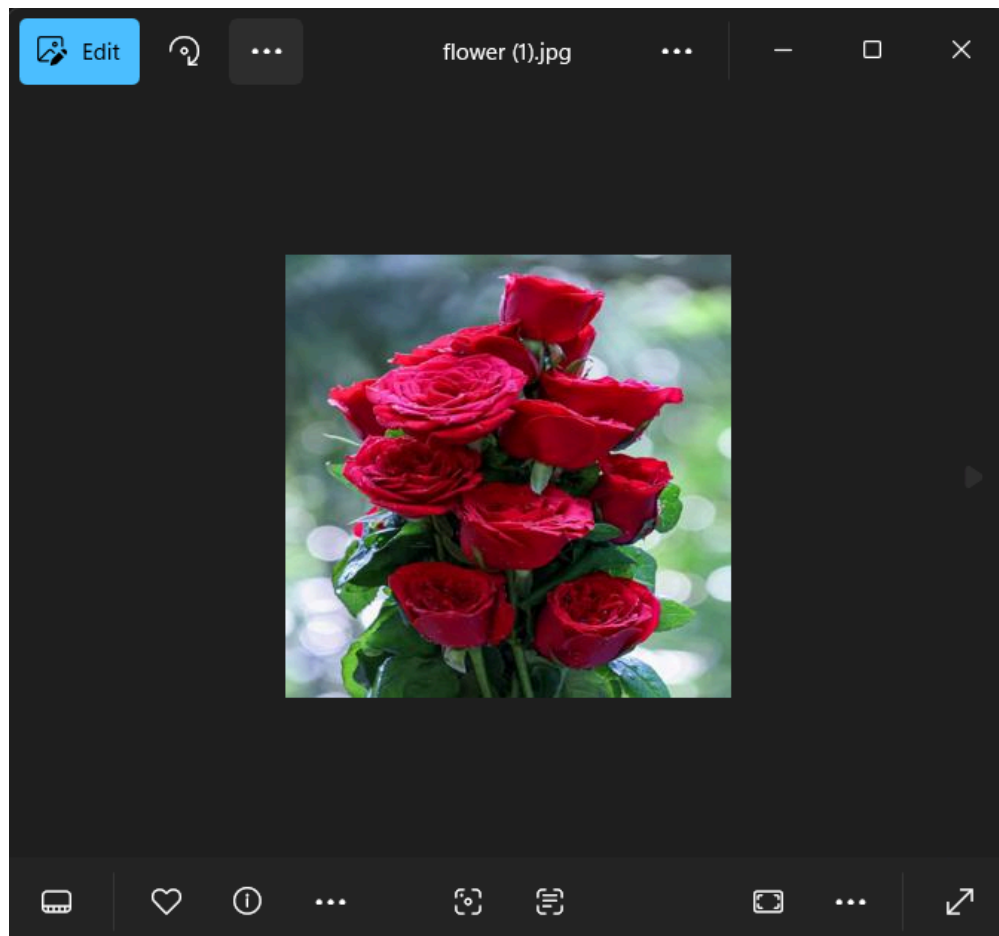


*Fig. 11 : CloudWatch Execution Logs*  
Shows successful Lambda execution, including the original image size and resized image size (300 × 300).



*Fig. 12 : API Response with Image URL*

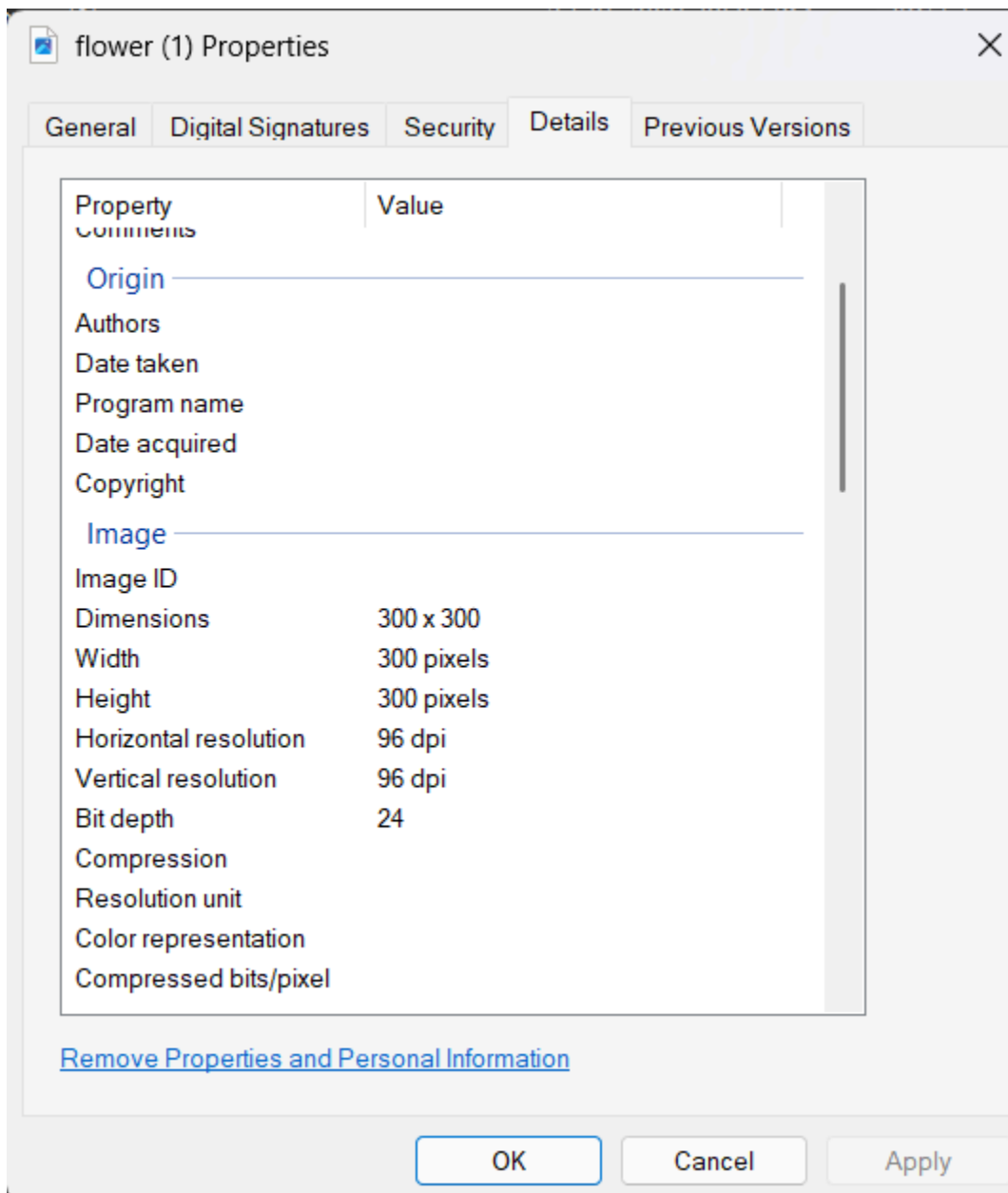
Displays the JSON response returned by API Gateway containing the processed image URL.



*Fig. 13 : Processed Image Output*

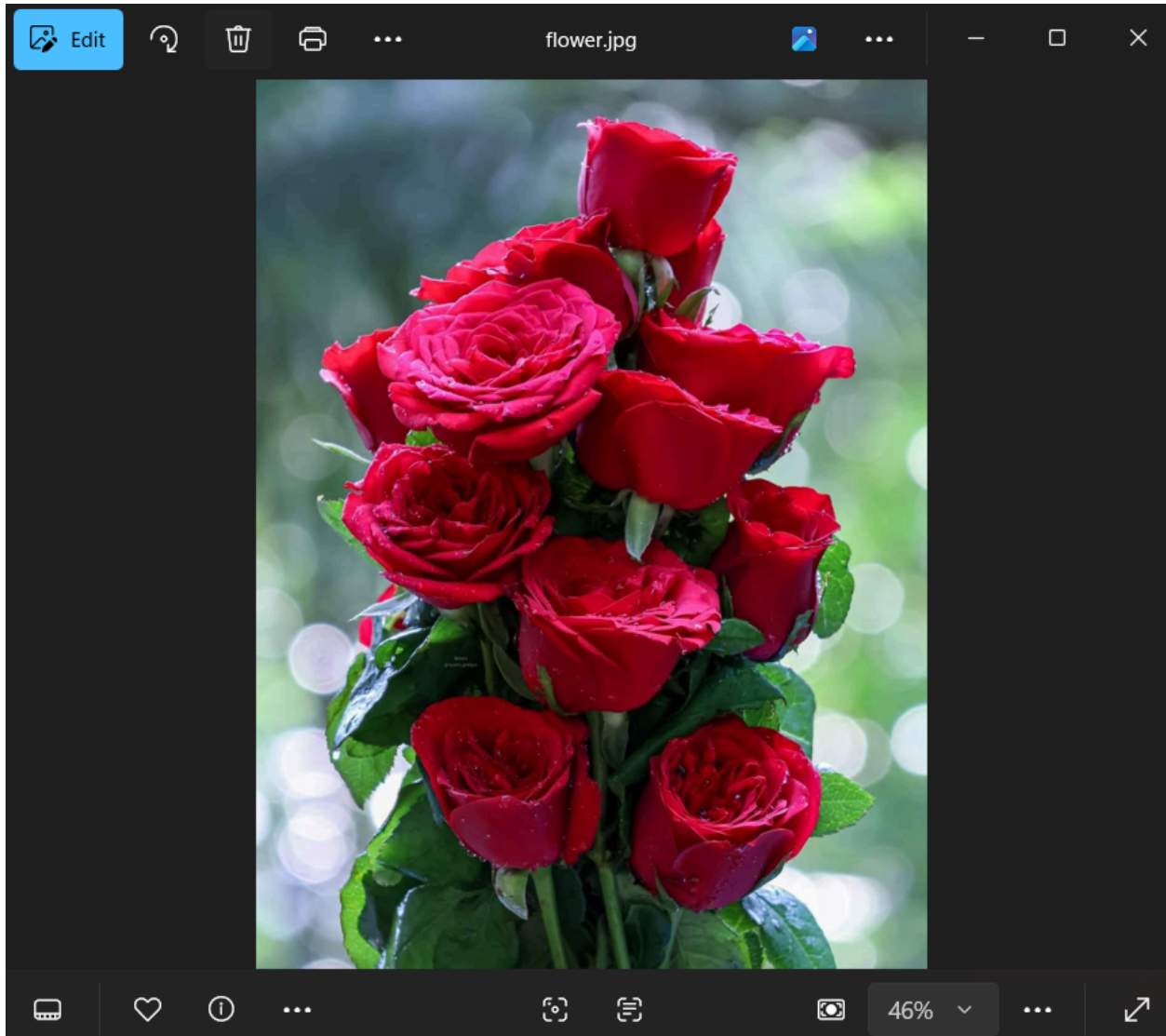
Shows the resized image opened successfully using the generated URL.





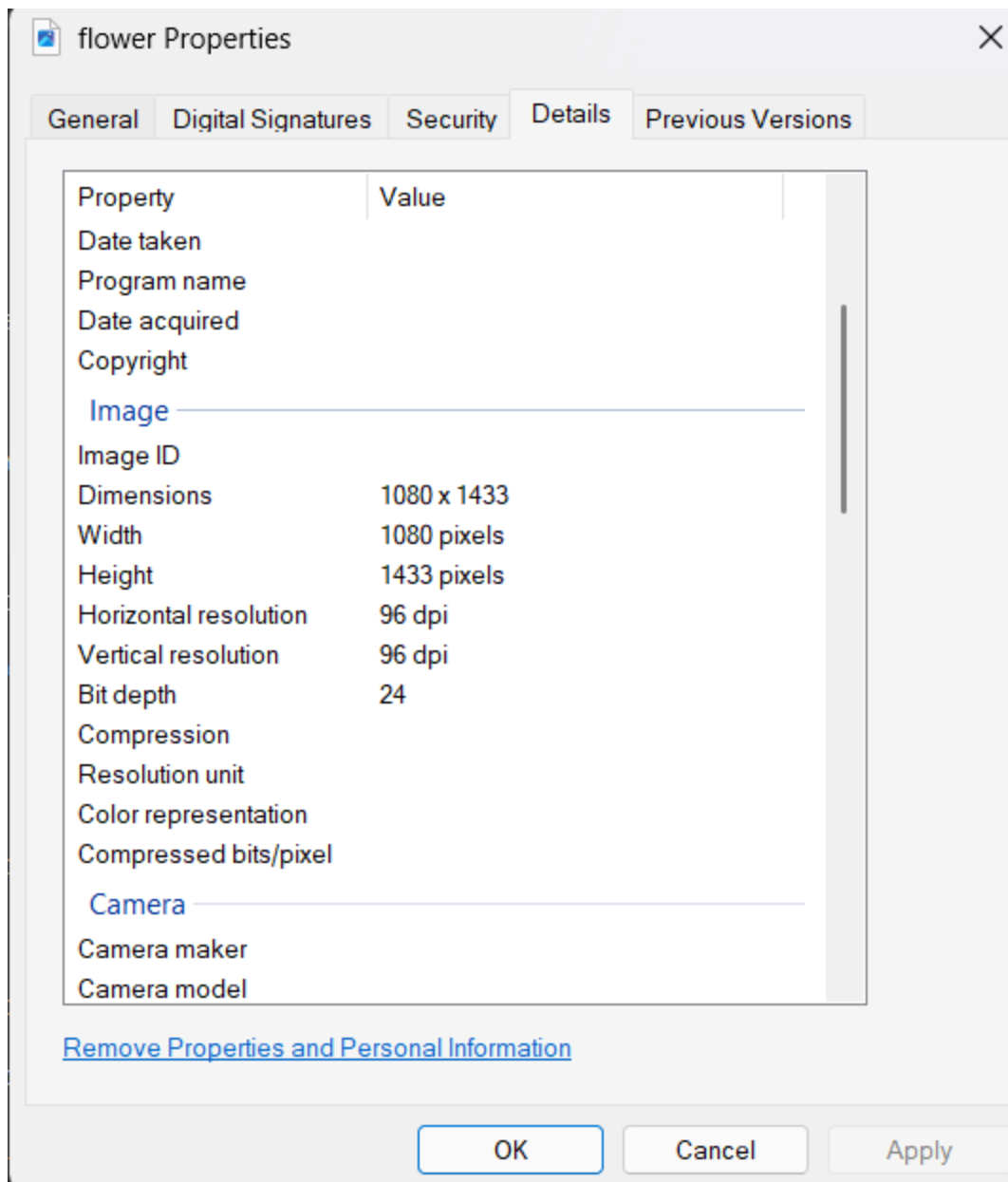
*Fig. 14 : Processed Image Properties*

Verifies that the processed image dimensions are **300 × 300 pixels**, confirming successful resizing.



*Fig. 15 : Original Image*

Displays the original uploaded image before image processing.



*Fig. 16 : Original Image Properties*

Shows the original image dimensions (**1080 × 1433 pixels**) before resizing.